

Experiment No. 4

Experiment Name

Advanced Data Visualization using Matplotlib and Seaborn

Aim

To create and analyze various data visualizations using Python libraries such as Matplotlib and Seaborn in order to identify trends, patterns, correlations, and insights from real-world datasets.

Purpose

Data visualization transforms raw data into graphical representations, making it easier to understand complex datasets and communicate insights effectively. Well-designed visualizations help analysts identify trends, outliers, distributions, and relationships that may not be apparent from numerical data alone.

In this experiment, students will use datasets such as the Superstore Sales Dataset, Netflix Movies and TV Shows Dataset, COVID-19 Dataset, or World Happiness Report to create meaningful visualizations. Students will also learn how to select the most appropriate chart based on the nature of the data and the analytical objective.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, Matplotlib, and Seaborn libraries
- Understanding of data preprocessing and Exploratory Data Analysis (EDA)
- Jupyter Notebook or Google Colab
- Cleaned dataset from previous experiments

Steps

1. Select a real-world dataset such as the Superstore Sales Dataset, Netflix Dataset, COVID-19 Dataset, or World Happiness Report.
2. Import the required libraries (Pandas, Matplotlib, and Seaborn) and load the cleaned dataset.
3. Create bar charts to compare categorical variables such as sales by region, genre distribution, or product categories.
4. Generate line charts to analyze trends over time, such as monthly sales, yearly movie releases, or daily COVID-19 cases.

5. Plot histograms to study the distribution of numerical variables such as sales, ratings, or customer age.
6. Create box plots to identify outliers and compare distributions across different categories.
7. Generate scatter plots to analyze relationships between two numerical variables, such as advertising expenditure vs. sales or budget vs. revenue.
8. Create a correlation heatmap using Seaborn to visualize relationships among numerical features.
9. Customize the visualizations by adding titles, axis labels, legends, annotations, and appropriate color palettes to improve readability.
10. Interpret each visualization and prepare a summary highlighting key business insights obtained from the graphical analysis.

Questions

1. Why is data visualization considered an essential component of data analytics?
2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?
3. Which type of chart would you choose to compare sales across different regions? Justify your answer.
4. What information can be obtained from a histogram and a box plot?
5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?
6. What is a correlation heatmap? How can it assist in feature selection for machine learning?
7. Why is chart customization (titles, labels, legends, colors) important in data visualization?
8. What factors should be considered while selecting an appropriate visualization for a dataset?
9. How can misleading visualizations affect business decision-making? Give a real-world example.
10. Explain how data visualization supports storytelling and business intelligence in organizations.